

Inhibition of biofilm formation and quorum sensing mediated virulence in *Pseudomonas aeruginosa* by marine sponge symbiont *Brevibacterium casei* strain Alu 1

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ABSTRACT

The alternative antimicrobial strategies that mitigate the threat of antibiotic resistance is the quorum-sensing inhibition (QSI) mechanism, which targets autoinducer dependent virulence gene expression in bacterial pathogens. N-acyl homoserine lactone (AHL) acts as a key regulator in the production of virulence factors and biofilm formation in *Pseudomonas aeruginosa* PAO1 and violacein pigment production in *Chromobacterium violaceum* ATCC 12472. In the present study, the marine sponge *Haliclona fibulata* symbiont *Brevibacterium casei* strain Alu 1 showed potential QSI activity in a concentration-dependent manner (0.5–2% v/v) against the N-acyl homoserine lactone (AHL)-mediated violacein production in *C. violaceum* (75–95%), and biofilm formation (53–96%), protease (27–82%), pyocyanin (82–95%) and pyoverdine (29–38%) productions in *P. aeruginosa*. Further, the microscopic analyses validated the antibiofilm activity of the cell-free culture supernatant (CFCS) of *B. casei* against *P. aeruginosa*. Subsequently, the biofilm and pyoverdine inhibitory efficacy of the ethyl acetate extract of *B. casei* CFCS was assessed against *P. aeruginosa*. Further, the gas chromatography–mass spectrometry (GC-MS) analysis revealed the presence of variety of components in which diethyl phthalate was found to be a major active component. This phthalate ester, known as diethyl ester of phthalic acid, could act as a potential therapeutic agent for preventing bacterial biofilm and virulence associated infectious diseases.

1. Introduction

The growing problem of microbial resistance to current antibiotics and high spreading ratio of resistant bacterial species has become a major public health concern and the biggest challenge faced by the global healthcare sector [1]. *Pseudomonas aeruginosa* is an aerobic, ubiquitous Gram-negative bacterium belonging to the family *Pseudomonadaceae* that is able to survive in a wide range of environments. It is considered as a foremost clinical concern due to its opportunistic nature to cause acute and chronic human infections such as cystic fibrosis, bronchiectasis, pneumonia and bacteraemia, particularly in immune-compromised patients [2,3].

P. aeruginosa is the third most common organism responsible for about 57% of all nosocomial infections [4,5]. The entire genome of *P. aeruginosa* (5.5–7 Mbp) is relatively large compared to other sequenced bacteria such as *Bacillus subtilis* (4.2 Mbp), *Escherichia coli* (4.6 Mbp) and *Mycobacterium tuberculosis* (4.4 Mbp). It encodes a large proportion of regulatory enzymes important for metabolism, transportation and efflux of organic compounds. This improved coding capability of the *P. aeruginosa* genome permits greater metabolic versatility and high adaptability to environmental changes [6,7]. The key pathways employed by *P. aeruginosa* to establish pathogenesis are; the production of an array of virulence factors and the formation of biofilm, providing the bacteria flexibility with survival in the host

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system. Biofilm formation is one of the mechanisms used by bacteria for developing resistance by chemical signaling to build protective structures against conventional antimicrobial agents [8,9]. Hence, eradication of cells in the biofilm matrix becomes extremely difficult. Cellular coordination is brought about by events of intercellular communication using small signaling molecules such as N-acyl homoserine lactone (AHL), which is collectively called Quorum Sensing (QS) [10]. Bacterial QS system depends on population density that regulates virulence genes expression and cellular differentiation to optimize their physiology for a particular environmental stimulus [11].

In Gram negative bacteria, the QS intervened pathogenesis occurs through AHL signal dependent mechanism [12]. AHL is known to play key role towards virulence in number of pathogenic bacteria including *P. aeruginosa*, *Serratia marcescens*, *Aeromonas hydrophila*, *A. salmonicida*, *Yersinia ruckeri*, *Hafnia alvei*, *Vibrio harveyi*, *V. vulnificus* and *V. anguillarum* [13]. Therefore, disrupting the AHL would be a useful target and powerful strategy to prevent QS dependent bacterial infections [14]. QS Network in *P. aeruginosa* consists of three QS systems including *lasIR*, *rhlIR* and PQS system that are hierarchically related to each other, and *lasIR* system is located at the top of this hierarchy [15, 16]. Also, MvfR/HAQ is the multiple virulence factor regulator acts as a transcriptional factor for 4-hydroxy-2-alkylquinoline (HAQ) system in *P. aeruginosa* QS mechanism. Synthesis of MvfR co-ligand is essential for MvfR activation whereby mutations in these systems led to the deficiency of pyocyanin synthesis [17]. These QS systems regulate virulence genes expression in *P. aeruginosa*. *lasIR* system regulates the genes responsible for *lasA* (protease), *lasB* (elastase), *tox A* (exotoxin A), whereas *rhlIR* regulates the genes for *rhlAB* (rhamnolipids), *aprA* (alkaline protease), *lecA* (lectins) and PQS regulates *phzABCDEFGHI*, *phzM* (pyocyanin). Products of all these genes help *P. aeruginosa* in the establishment of infections. In view of the background it is necessary to identify novel compounds which can disrupt the communication among the bacteria by impairing QS [18].

QS controls the virulence factors and biofilm formation for the development of acute and chronic infections in *P. aeruginosa*. These factors result in bacterial persistence and reduced sensitivity to antimicrobials [19]. *P. aeruginosa* uses an array of virulence factors and numerous mechanisms, including biofilm formation, to successfully infect a diverse range of hosts and to defend itself from environmental stress and antibiotics [20–22]. Thus, anti-QS strategies could be a prospective target to prevent *P. aeruginosa* infection. Hence, there is an increasing need to search potent anti-QS and antibiofilm compounds from the natural resources.

Marine sponges belong to the Phylum *Porifera* and are among the most primitive metazoans that manufacture diverse type of bioactive metabolites. They are of great ecological significance due to their immense capability to filter enormous volumes of seawater. Sponges are well known to harbour assorted microbes and represent a substantial source of bioactive natural compounds derived from the marine environment. Recent studies of the microbial communities of marine sponges have exposed untapped and unexplored plethora of species and an array of new chemical compounds with antimicrobial, antitumor, anti-inflammatory, mosquitocidal, insecticidal and immunosuppressive activity [23–25]. Similarly, Manoalide derivatives in the extract of marine sponge *Luffariella variabilis* have shown a strong QS inhibitory (QSI) activity and acted as a good antagonist against the QS systems of *P. aeruginosa* [26]. Microbial symbionts especially actinomycetes in marine sponges are known for their unprecedented ability to produce novel lead compounds of clinical and pharmaceutical importance and serve as marine natural product factory that sustainably supplies bioactive compounds. Although the marine sponges are known for their bioactive potential, studies on its QSI properties remain scanty. Hence, the present study was aimed to evaluate the anti-QS and antibiofilm activity of marine sponge associated *Brevibacterium casei* against *P. aeruginosa*.

2. Materials and methods

2.1. Isolation of marine sponge symbionts

The marine sponge was collected approximately 10–15 km from the shore and at the depth of 10–15 feet in the Palk Strait region (Lat 9°44' N; Long 79°018' E), Tamil Nadu, India. Sample was collected in sterile polythene bag and transported to the laboratory within minimum possible time to avoid the external microbial contamination and excessive proliferation. The pieces were unbound from adhering particles by vortexing twice for 20 s with 2 ml of sterile seawater. The seawater was decanted, which was once again placed with sterile seawater with continued vortexing between washings [27].

One milliliter of homogenized sponge sample was mixed with 9 ml of sterile water blank, respectively. This suspension was serially diluted up to 10^{-6} dilution. 1 ml of the diluted sample was taken from 10^{-3} and 10^{-4} dilutions and was pour plated with 15–20 ml Yeast Malt Agar (YM Agar – Himedia-M4242) (ISP Medium No. 2) and incubated at room temperature ($28 \pm 2^\circ\text{C}$) for 5 days. To eliminate the bacterial and fungal contamination, 8 ml of 1% Nystatin was added to 1000 ml of the sterilized medium [27].

2.2. Microbial strains, culture media and conditions

Actinomycetes from sponge sample were isolated on YM Agar (HiMedia Laboratories, Mumbai, India) [28]. *Chromobacterium violaceum* ATCC 12472 was used as a QS biomonitor organism. The strain *P. aeruginosa* PAO1 (ATCC 15692) was gifted by Prof. Jo Handelsman (Wisconsin University), which was grown on Luria Bertani (LB) agar plates at 27°C for control the swarming characteristic of PAO1 and the stock culture was maintained at -80°C in 25% glycerol. For each experiment, pure single colony was inoculated into fresh LB broth and incubated at 37°C for 24 h.

2.3. Preliminary analysis by using marker strain

Agar overlay assay was used to study the anti-QS activity of marine sponge associated actinomycetes by following the method of Thenmozhi et al., [29]. *Chromobacterium violaceum* biosensor method was used in this study [30]. 10 ml of warm molten soft agar (LB with 0.5% agar) was seeded with 10 μl of 18 h cultured *C. violaceum* and mixed well. Then, the mixed culture was poured over the surface of a solidified YM Agar plate containing actinomycete colony to form the overlay. The assay plates were incubated at 37°C for 24 h and observed for the presence of a colourless, opaque, and halo zone around the colony was considered as anti-QS activity of the test sample. LB plate containing *C. violaceum* was used as control.

2.4. Viocaine inhibition assay in *C. violaceum*

The filter-sterilized cell-free culture supernatant (CFCS) of active lead actinomycetes strain was analysed for their QSI potential against *C. violaceum* through qualitative and quantitative analysis. After 24 h of incubation, the actinomycete culture was centrifuged at 8000 rpm for 10 min and the CFCS was filtered through a 0.22 μm membrane filter (Millipore, India). An amount of 1% of *C. violaceum* culture (OD 0.4 at 600 nm) was inoculated in sterile 24-well microtitre plates (MTPs) containing 1 ml of LB broth supplemented with and without active actinomycete CFCS, at concentrations of 1%, 2%, 3%, 4% and 5% v/v and then incubated at 30°C for 24 h. After incubation, the growth OD was taken at 600 nm. Then, 1 ml of the culture was centrifuged at 8000 rpm for 10 min. The pellet was resuspended in 1 ml of dimethyl sulfoxide (DMSO). The suspension was then vortexed vigorously to extract the violacein pigments and then centrifuged at 10,000 rpm for 10 min. The absorbance of supernatant was measured spectrophotometrically at 585 nm (UV-visible spectrophotometer; Shimadzu, UV-2450, Japan) [31].

Inhibition of violacein production was calculated in terms of percentage of inhibition by using the following formula:

$$\text{Percentage of inhibition} = ((\text{control OD}_{585\text{nm}} - \text{test OD}_{585\text{nm}}) / \text{control OD}_{585\text{nm}}) \times 100$$

2.5. Identification of active actinomycetes isolated from marine sponge and their phylogenetic analysis

Total genomic DNA and 16S rRNA gene amplification of active marine sponge associated actinomycete was done as described previously [32]. Sequencing of the 16S rRNA gene (about 1000 bp) was done in Macrogen (Seoul, Korea). The sequences obtained were assembled, analysed and manually edited using CAP3 software package and compared to sequences within the NCBI database (<http://www.ncbi.nlm.nih.gov/>) using BLAST as described previously [33]. Multiple sequence alignment and Neighbor Joining plot were carried out using CLUSTALX. Bootstrap analysis was performed to validate the reproducibility of the branching pattern. The 16S rRNA sequence of marine actinomycete was identified as *Brevibacterium casei* strain Alu 1 (Genbank accession number MN186823.1). Further, the QSI potential of *B. casei* CFCS was used against the production of biofilm and other virulence factors of *P. aeruginosa*.

2.6. Effect of *B. casei* CFCS on biofilm biomass, pyocyanin, protease and pyoverdinin quantification assays

The effect of *B. casei* CFCS on the biofilm formation of *P. aeruginosa* was determined by quantifying the biofilm biomass. 10 µl of *P. aeruginosa* cell suspension with 0.4 OD at 600 nm (1×10^8 CFU/ml) was added to 1 ml of fresh LB medium in the presence (1% and 2% v/v) and absence of actinomycete CFCS in the wells of MTP and incubated statically at 28 °C for 24 h. After incubation, the planktonic cells in MTP wells were removed by rinsing the wells twice with distilled water. The surface attached cells were stained with 1 ml of 0.2% crystal violet (CV) solution (HiMedia, India) for 5 min followed by washing with distilled water and destaining with 20% glacial acetic acid (GAA). The biofilm biomass was quantified by the intensity of dissolved CV using UV-Visible spectrophotometer at OD 570 nm [34].

Briefly, pyocyanin pigment was extracted from CFCS of overnight grown *P. aeruginosa* culture treated with 0.5%, 1% and 2% v/v concentrations of actinomycete extract using equal volume of chloroform and re-extracted using 0.2 N HCl to get a pink to deep red coloured solution. The absorbance of the solution was measured at 520 nm and inhibition was calculated as compared to control [35].

The total proteolytic activity of *P. aeruginosa* was determined by the azocasein assay. 150 µl of both extract treated and untreated culture supernatants were added to 500 µl of 0.3% azocasein (Sigma, St. Louis, MO, USA) in 0.05 M Tris-HCl, 0.5 mM CaCl₂ (pH 7.5) and incubated at 37 °C for 30 min. Enzymatic reaction was stopped by adding ice cold trichloroacetic acid (10%, 0.5 ml) and the reaction was kept at 4 °C for 10 min. After incubation, the insoluble azocasein was removed by centrifugation at 10,000 rpm for 10 min and the obtained supernatant was measured at OD 400 nm [26].

Moreover, pyoverdinin, the major siderophore produced by *P. aeruginosa*, is a signaling molecule to regulate the production of several secreted toxins [36,37]. For quantify the pyoverdinin production, *P. aeruginosa* was cultivated in the presence and absence of *B. casei* CFCS. The relative concentration of pyoverdinin was measured at an excitation wavelength of 405 nm and an emission wavelength of 465 nm [1], which directly represents the fluorescence intensity of the culture supernatant.

2.7. Hydrophobicity index

The effect of *B. casei* CFCS on the cell surface hydrophobicity (CSH) of test pathogens was determined through bacterial adhesion to hydrocarbons (BATH) assay as a measure of their hydrophobic adherence to toluene by following the method of Santhakumari et al., [38]. Briefly, 1 ml of untreated control and *B. casei* CFCS (1% and 2% v/v) treated *P. aeruginosa* culture was taken in glass tubes and an equal volume of toluene was added. Then, the mixture was vigorously vortexed for 2 min and incubated for 10 min at room temperature to allow separation and then the OD at 600 nm of aqueous phase was measured. Bacterial culture incubated with toluene and without CFCS acted as the control. The percentage of hydrophobicity index was calculated by the following formula:

$$\% \text{ of hydrophobicity index} = [1 - (\text{Final OD after vortexing} / \text{Initial OD before vortexing})] \times 100.$$

2.8. EPS quantification, Congo red binding and rhamnolipid assay

Extracellular polymeric substances (EPS) of *B. casei* CFCS treated and untreated *P. aeruginosa* samples were extracted by phenol-sulphuric acid method as described by Sivasankar et al. [39], with slight modification. Briefly, *P. aeruginosa* was grown in absence and presence of *B. casei* CFCS (1% and 2% v/v) for 18 h at 28 °C in 24 well MTP. After incubation, the planktonic cells were washed out by sterile distilled water. Then the cells were dissolved by 0.9% NaCl (1 ml) and equilibrated phenol (1 ml). Then, 5 volume of H₂SO₄ was added to mix and incubated in dark at room temperature for 1 h. Then the absorbance was taken at 490 nm. The percentage of EPS inhibition was calculated by using the following formula.

$$(\text{Control}_{\text{OD}} - \text{Treated}_{\text{OD}}) / \text{Control}_{\text{OD}} \times 100$$

Production of EPS was also estimated by the Congo red binding method of Goswami et al. [40], with minor modification. Biofilms were grown as mentioned earlier; subsequently actinomycete extract was added at test concentrations (1% and 2% v/v) to the developed biofilms and incubated for 24 h at 37 °C. Planktonic cells were discarded and the adhered cells were washed with phosphate-buffered saline (PBS). Congo red (1% w/v) was added to each well and following incubation in the dark for 30 min the extra dye was discarded. DMSO (200 µl) was added to each well to solubilise the bound dye and the absorbance was measured at 490 nm in a plate reader.

Rhamnolipid production was determined by the orcinol method as described earlier [41]. Briefly, the CFCS of both untreated and treated *P. aeruginosa* culture were extracted using diethyl ether. Then the extracts were centrifuged at 10,000 g for 8 min at 4 °C and the pellets were dissolved in 100 µl of distilled water. The resulting solution was mixed with 100 µl of 1.7% orcinol and 800 µl of 60% (v/v) sulphuric acid. The mixture was heated at 80 °C in a water bath for 30 min and then cooled down to room temperature. The absorbance was measured at 420 nm and the production of rhamnolipid was evaluated as the ratio between the Absorbance_{treated} and the Absorbance_{control}.

2.9. Pellicle formation assay

Pellicle formation was assessed as described earlier [42]. Concisely, 5 ml of LB containing 0.5% glucose were inoculated with *P. aeruginosa* in glass tubes (18 mm × 150 mm, BorosilIR, Borosil Glass Works Ltd., Kolkata, India). The 1% (v/v) of *B. casei* CFCS were added to the treatment set and the same proportions of DMSO were added to the control sets. The tubes were incubated vertically at 37 °C without any mechanical shaking and the pellicle formation at the air-liquid interface was visually observed after 72 h.

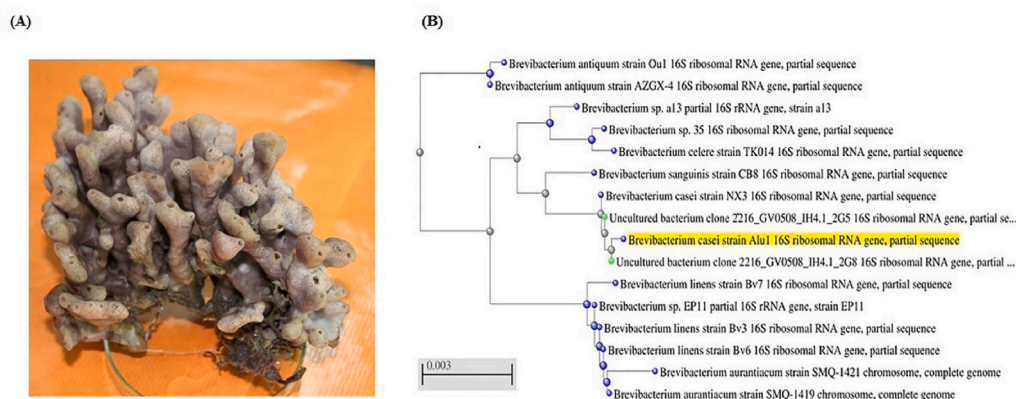


Fig. 1. (A) Photographic image of identified marine sponge *Haliclona fibulata* and (B) Phylogenetic tree construction of marine sponge symbiont *B. casei* using neighbor joining method.

2.10. Motility assessment

For swarming assay, 1% of *P. aeruginosa* was spot inoculated on the surface of semi-solid agar medium (1% peptone, 0.5% NaCl and 0.5% agar). Then the plates were incubated at 37 °C for 24 h [43].

2.11. Microscopic analysis

The antibiofilm potential of *B. casei* CFCS validated by microscopic analyses were done in accordance with Subramenium et al., [44]. Briefly, *P. aeruginosa* was allowed to form biofilms on glass slides (1 × 1 cm) in the presence and absence of *B. casei* CFCS at 2% v/v and incubated for 24 h at 37 °C. After incubation, the slides were washed with sterile PBS. Then the slides were stained using 0.4% CV or 0.1% acridine orange for Light microscopy (LM) (Nikon Eclipse 80i, Shinagawa-ku, Tokyo, Japan) and confocal laser scanning microscopy (CLSM) (Zeiss LSM 710, Carl Zeiss, Germany) analysis, respectively.

For field emission scanning electron microscope (FESEM) analysis, the slides were fixed using 2% glutaraldehyde for 1 h at 4 °C. This was followed by dehydration for 2 min with ethanol at increasing concentrations (50%, 60%, 70%, 80%, 90% and 100%). The fixed slides were sputter coated with gold prior to observation under FESEM (Quanta FEG 250, Thermo Fisher Scientific; Hudson, NH, USA).

2.12. Thermal stability of *B. casei* CFCS

The thermal stability of *B. casei* CFCS was assessed by exposing it to different temperatures such as overnight heating at 50 °C and autoclaving at 121 °C. The untreated CFCS was considered as the control. The heat treated CFCS (1% v/v) was subjected to pyocyanin inhibition assay by adding it to 1 ml of LB in MTP inoculated with *P. aeruginosa* (1%; OD 0.4 at 600 nm) and incubated at 30 °C for 24 h [45].

2.13. Solvent extraction of *B. casei* CFCS

The marine actinomycete *B. casei* was cultured in yeast malt agar (HiMedia Laboratories, Mumbai, India) and incubated for 48 h at 27 °C. The cells were pelleted, and the culture supernatant was filtered through a 0.2 µm filter. The cell free supernatant was successively extracted with solvents (250 ml) of different polarities, such as petroleum ether, dichloromethane, chloroform, ethyl acetate, and methanol, filtered using Whatman No.1 filter paper (Millipore, India). The filtrates were concentrated to dryness under reduced pressure in rotary vacuum concentrator (Christ, RVC 2–18) and stored at –20 °C [46]. Further, the successive extracts were used for assessment of QS inhibition properties against *P. aeruginosa* qualitatively. Yield of the extracts were calculated using the following formula:

$$\text{Yield(\%)} = \frac{\text{Wt. of the beaker with extract} - \text{Wt. of the empty beaker}}{\text{Wt. of the sample in grams}} \times 100$$

The solvent extracts were combined and evaporated to dryness under reduced pressure at room temperature to yield crude extracts, and each crude extract obtained was weighed [33]. The crude extracts were dissolved in methanol and screened for their QSI activity against *P. aeruginosa* through pyocyanin assay.

2.14. Effect of ethyl acetate extract of *B. casei* CFCS on pyoverdine and biofilm inhibition assay

Pyoverdine and biofilm productions in *P. aeruginosa* upon ethyl acetate extract of *B. casei* CFCS treatment (10, 20, 30, 40 and 50 µg/ml) were assessed by the method of Musthafa et al. [35], and Adonizio et al. [1], as described above.

2.15. GC-MS analysis of *B. casei* CFCS ethyl acetate extract

GC-MS analysis for crude *B. casei* CFCS ethyl acetate extract was carried out using Agilent GC 7890A, containing Agilent DB5MS gas chromatograph column (ID 0.25 mm, 0.25 µm thickness), coupled to a mass spectrometer mass detector (MS5975C); the oven temperature was set at 110 °C with 2 min hold, then increased to 300 °C at the rate of 10 °C/min with 2 min hold. The injector temperature was 250 °C. The injection volume was 1 µl. Injection mode was splitless, the pressure was 7.652 psi, and the total flow was 54 ml min⁻¹. Purge flow was 50 ml min⁻¹ and the split ratio was 100:1. The total run time was 28 min. The total ion chromatogram acquired was autointegrated by Chemstation and the constituents were identified by comparison with the mass spectral reference database of NIST library (2014) software (National Institute of Standards and Technology, Gaithersburg, USA).

2.16. Statistical analysis

All the experiments were performed in triplicate to validate reproducibility and the *P* values were calculated statistically by Student's *t*-test.

3. Results and discussions

Antibiotics that aim at controlling the growth of microorganisms have been applied in animal and human health and particularly in the treatment of bacterial infections. This approach has a noticeable drawback, when growth of bacteria is blocked, the bacteria are under harsh

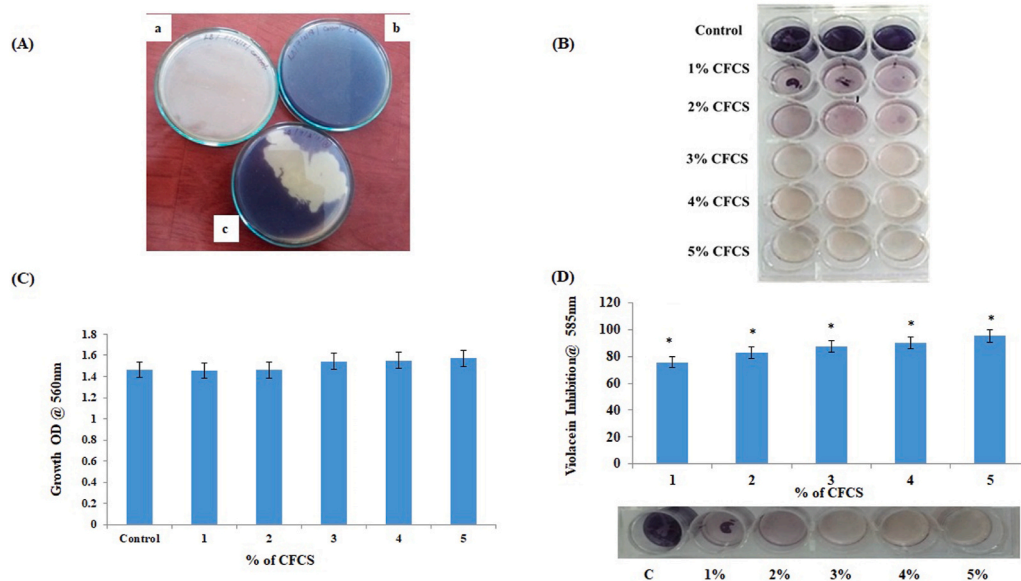


Fig. 2. (A) Analysis of the anti-QS activity of *B. casei* culture using the biomarker strain *C. violaceum* (a) blank, (b) control, (c) *B. casei* spotted plates. (B) Effect of *B. casei* CFCS on *C. violaceum* violacein pigment production. (C) Effect of *B. casei* CFCS on *C. violaceum* growth (D) Quantitative analysis of violacein production in *C. violaceum* upon *B. casei* CFCS treatment. Data represent mean values ($n = 3$) $\pm$ standard deviation. * indicates significance at $P < 0.005$.

selective pressure to develop resistance. So as the results, more and more drug-resistant bacteria come out, and the antibiotics lost their effects due to improper and uncontrolled uses of antibiotics against pathogenic bacteria which become a major health problem worldwide [1,47,48].

The discovery and explication of bacterial QS systems have been paralleled by growing awareness in the ability to manipulate signal reception and transduction. The pathogens are central regulators for the expression of virulence factors and represent highly attractive targets in the QS system for the development of novel therapeutics. QSI can competitively inhibit QS signaling system, providing an opportunity to develop new drugs against these targets to combat against virulence [49]. The *LasIR* and *RhlIR* systems direct the expression of several genes in *P. aeruginosa*, including production of pyocyanin pigment, virulence factors such as *LasA* staphylolytic protease, *LasB* elastase, and biofilm formation [1,30,50].

3.1. Isolation, identification and screening of marine sponge symbionts

In the current investigation, marine sponge was collected and identified as *Haliclona fibulata* (Fig. 1A) by Dr. P.A. Thomas, Vizhingan Research centre of Central Marine Fisheries Research Institute, Kerala, India. Then, a total count of 23 marine sponge associated actinomycetes were isolated and screened for their QSI activity against *C. violaceum* ATCC 12472 (Data not shown). Among them, the strain *B. casei* (Genbank accession No. MN186823.1) showed constant QSI activity (Fig. 2A). The phylogenetic tree of isolated active QSI strain *B. casei* was shown in Fig. 1B. Further, *B. casei* CFCS was examined to analyze its efficacy to interfere with *P. aeruginosa* QS regulated virulence factors production such as protease, pyocyanin, pyoverdine, rhamnolipid, EPS, pellicle formation, motility and biofilm formation.

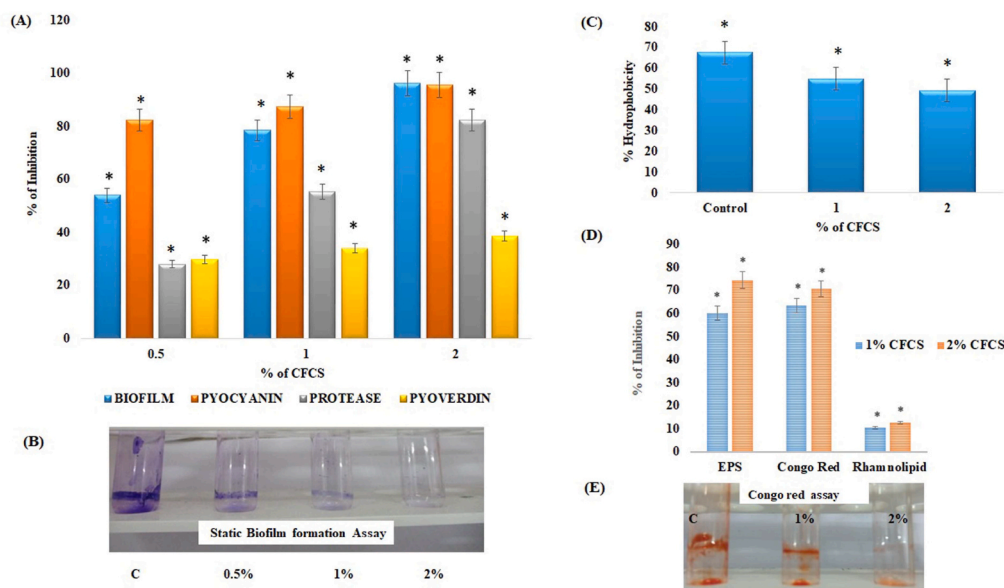


Fig. 3. Effect of *B. casei* CFCS on *P. aeruginosa* (A) biofilm, pyocyanin, protease and pyoverdine productions, (B) static ring biofilm formation, (C) cell surface hydrophobicity index, and (D) EPS (quantified through phenol-sulphuric acid and Congo red assay) and rhamnolipid productions (D). Data represent mean values ($n = 3$) $\pm$ standard deviation. * indicates significance at $P < 0.005$. (E) Illustrative image of test tubes displaying EPS inhibitory (Congo red assay) activity of *B. casei* CFCS on *P. aeruginosa*.

3.2. Qualitative and quantitative analysis of violacein inhibition

Biosynthesis of violacein, a purple pigment regulated by CviR QS system in the biomarker strain *C. violaceum* confirmed the QSI property. In qualitative violacein inhibition assay, 1%, 2%, 3%, 4% and 5% v/v CFCS of *B. casei* isolated from marine sponge *Haliclona fibulata* was assessed for its ability to inhibit AHL-mediated violacein production in *C. violaceum*. Biosynthesis of violacein can interfere with CviR–AHL complex and prevent the expression of *vio* genes that regulates violacein production [51]. The outcome of this study revealed that the CFCS of *B. casei* exhibited a significant anti-QS activity against *C. violaceum* (Fig. 2B). In which, a minimum of 75.7% and a maximum of 95.3% violacein inhibition was observed at 1% and 5% CFCS (v/v) treated *C. violaceum* samples, respectively (Fig. 2C and D). This result clearly shows that the negative effect of *B. casei* CFCS on pigment production was due to disruption of the QS mechanism without disturbing bacterial growth. Interference of such QS mechanism in *C. violaceum* by coral-associated bacteria (CAB) *Stenotrophomonas maltophilia*, rhizosphere soil bacterium *B. subtilis* R-18 and *Delftia tsuruhatensis* extract isolated from the rhizosphere of *C. laevigatus* have shown violacein inhibition which has been reported earlier [45,52,53]. However, it is difficult to compare the results with previous reports because of variation in the extraction methods and other parameters.

3.3. Effect of *B. casei* CFCS on *P. aeruginosa* biofilm, pyocyanin, protease and pyoverdinin production

The antibiofilm activity of *B. casei* CFCS against *P. aeruginosa* was assessed by biofilm biomass quantification assay. The obtained results demonstrated a dose-dependent biofilm reduction in *P. aeruginosa* upon *B. casei* CFCS treatment (Fig. 3A). At 2% (v/v) concentration, *B. casei* CFCS showed a maximum of 96% biofilm inhibition against *P. aeruginosa*, which is also evinced through static ring biofilm assay (Fig. 3B). Further, the microscopic analysis (LM, CLSM and SEM micrographs) demonstrated a significant reduction in the formation of microcolonies and multilayer biofilms due to the massive reduction in the quantity of EPS validated the antibiofilm efficacy of *B. casei* CFCS (2% v/v) against *P. aeruginosa* (supplementary file).

Furthermore, AHL-dependent QS inhibition by AHL for receptor binding and eventually inhibit biofilm formation of *P. aeruginosa*. However, no significant effect on growth of *P. aeruginosa* was observed in the presence of above *B. casei* CFCS was also analysed using the *C. violaceum*. We speculated that the active substance was associated with AHL analogs that proceeded as AHL-antagonists by competing with concentration of *B. casei* CFCS. Recent findings of Devi et al. [45], and Ma et al. [54], revealed that *B. subtilis* R-18 and *Staphylococcus hominis* extract showed a significant reduction in biofilm biomass against *S. marcescens* and *P. aeruginosa*, respectively. The biofilm architecture in *B. casei* CFCS treated slides was diminished and poorly developed when compared with that of the untreated control. With this observation, *B. casei* CFCS treatment is envisaged to control the biofilm-related infections caused by *P. aeruginosa*.

Pyocyanin, a blue phenazine pigment contributes to the persistence of *P. aeruginosa* infection by causing harmful effects toward lung [55] epithelial cells and by releasing inflammatory response initiated by the host. Pyocyanin synthesis is controlled by a complex synchrony of *lasR-lasI*, *rhlR-rhII* and *myfR-haq* QS-system whereby mutations in these systems led to the deficiency of pyocyanin synthesis [56]. Also, it can be reduced by disrupting the *aro* pathway which is responsible for the synthesis of chorismic acid from shikimate. Because, chorismic acid is the precursor of pyocyanin.

In order to analyze the efficiency of QS-dependent pyocyanin production, the PAO1 cells were cultivated in the presence and the absence of *B. casei* CFCS. A significant decrease in pyocyanin production of *P. aeruginosa* was observed to the level of 82.39%–95.6% after treatment at 0.5–2% v/v concentration (Fig. 3A). Moreover, a significant reduction

in the pyocyanin production was observed when *P. aeruginosa* was treated with the extract of edible fruits [33] and marine *Bacillus* sp. SS4 and phenylacetic acid [57,58]. From our studies and available literature, it could be postulated that *B. casei* CFCS may have inhibited the QS-systems at its gene expression level or acts as an antagonist to decrease pyocyanin production.

The proteases *LasA* and *LasB* of *P. aeruginosa* are considered to be one of the major virulence factors play vital role during acute infection and cause host tissue degradation [59]. In the present study, the ability of *B. casei* CFCS in reducing QS-dependent azocasein degrading protease activity was observed at 0.5%, 1% and 2% v/v in the supernatant of *B. casei* treated *P. aeruginosa*, with that of untreated *P. aeruginosa* supernatant was found to be 27.99%, 55.19% and 82.37%, respectively as shown in Fig. 3A. Moreover, the test extract showed no considerable growth inhibitory activity against *P. aeruginosa* at the tested concentration. The findings of Annapoorani et al. [26], Padmavathi et al. [60], and Singh et al. [61], have showed protease inhibition in *P. aeruginosa* and *S. marcescens* from various sources. The obtained results of this study showed that the CFCS of *B. casei* inhibits the cell-to-cell communication and thus prevents development of disease, including secreted factors such as proteases in *P. aeruginosa* from developing a robust biofilm.

In the mode of pathogenesis of *P. aeruginosa*, the pyoverdins are fluorescent siderophores which competes directly with mammalian transferrin for iron, and acts as an essential element for the virulence to host cells by promoting the pathogenicity by stimulating bacterial growth [62]. In this study, a dose-dependent pyoverdinin reduction was observed upon *B. casei* CFCS treatment whereas 0.5%, 1% and 2% (v/v) of CFCS showed 29.74%, 33.95% and 38.52% of inhibition, respectively (Fig. 3A). The observed results are in good agreement with the findings of Adonizio et al. [1], and Nithya et al. [28], wherein the marine bacterial strain *Bacillus pumilus* and medicinal plant extracts such as *Conocarpus erectus*, *Callistemon viminalis*, *Bucida buceras* exhibit potential antibiofilm and antivirulence activity against *P. aeruginosa*. Due to the potential inhibitory activity, *C. erectus*, *C. viminalis*, *B. buceras*, and the marine bacterial strain *B. pumilus* are considered as a positive control in the present study.

3.4. Hydrophobicity index

The hydrophobicity of bacterial pathogens contributes to their adherence, which is responsible for biofilm formation and adhesion to epithelial cells. The adherence of bacterial cells to the epithelial surface is considered to be an important factor in the development of infection [63]. Our findings revealed that the level hydrophobicity index in *B. casei* CFCS treated PAO1 was 54.98 and 49.23% at 1% and 2% (v/v) concentrations, respectively and the level hydrophobicity index of untreated control is 68% (Fig. 3C). As expected, the CFCS of *B. casei* effectively reduced the production of hydrophobicity index of PAO1 compared to the untreated control. Therefore, the antibiofilm activity of *B. casei* CFCS could be due to the inhibition of hydrophobicity index and EPS synthesis. Our finding supports the previous report made by Srinivasan et al. [64], who reported that the antibiofilm activity of silver nanoparticles against uropathogens *S. marcescens* and *P. mirabilis* strains was due to the hydrophobicity index and EPS inhibitions. Similarly, Santhakumari et al. [34], detected that the effect of 2, 6-Di-*tert*-butyl-4-methylphenol on hydrophobicity of *Vibrio* spp. was reduced significantly.

3.5. EPS quantification, Congo red binding and rhamnolipid assay

Microbial cells cocooned themselves in self-secreted EPS that one of the major virulence factors which is mainly involved in biofilm formation and establishing the infection in host tissue. It also prevents the penetration of antibiotics substance into the bacterial cell [65]. In the present study, *B. casei* CFCS significantly inhibited EPS production of

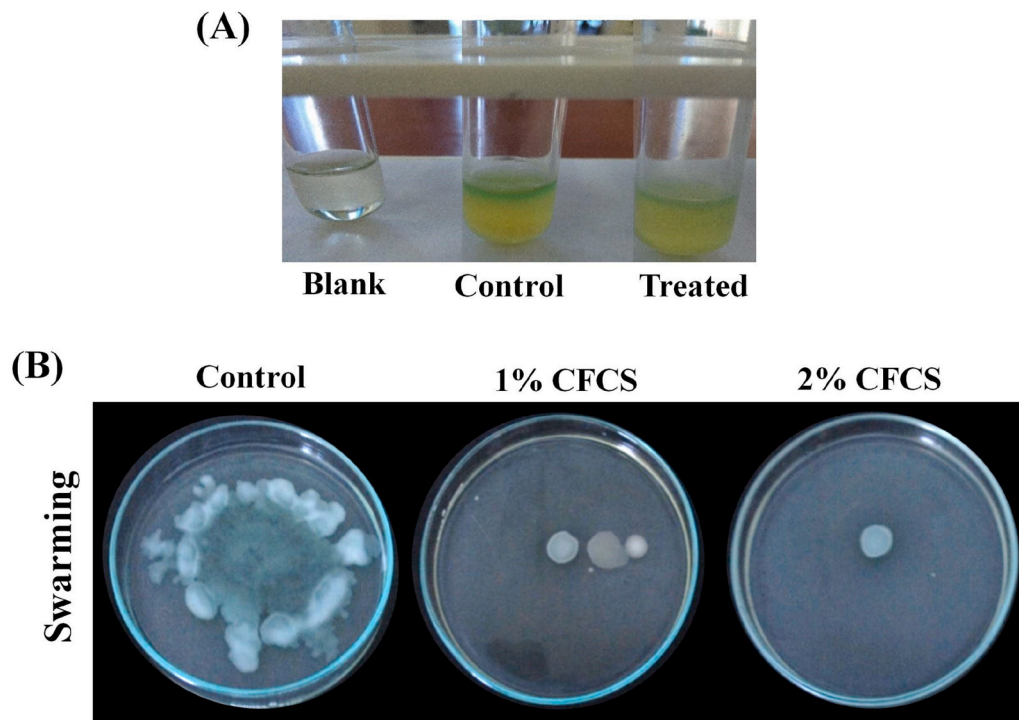


Fig. 4. (A) Effect of *B. casei* CFCS on *P. aeruginosa* pellicle formation, and (B) swarming motility.

P. aeruginosa to the level of 60% and 74.34% at 1% and 2% v/v concentrations, respectively (Fig. 3D). In support to our findings, Sankar Ganesh and Ravishankar Rai [66] have reported that the methanol extract of *Terminalia bellerica* reduced the EPS production of *P. aeruginosa* PAO1 when compared to that of untreated control. Moreover, *B. casei* CFCS significantly inhibited EPS production of *P. aeruginosa* might permit the penetration of antibiotics to this bacterium. Therefore, it could be used for the treatment of *P. aeruginosa* infections that involve biofilm formation.

Congo red is a dye which binds to the biofilm EPS [67]. The Congo red binding assay was done to ascertain the effect of *B. casei* CFCS on EPS production by *P. aeruginosa*. *B. casei* CFCS reduced the aggregation of EPS and thus, a lower amount of Congo red could bind to the treated biofilm compared to the control set. The amount of Congo red in the supernatant was significantly decreased from 100% (control) to 63.42% and 70.58% in the 1% and 2% v/v treatment set, respectively (Fig. 3D). A detachment assay was performed to observe whether *B. casei* CFCS had any effect on biofilm adhesion to the substratum. The CFCS exhibited a change in the adhesion ability of the biofilm cells (Fig. 3E). The number of planktonic cells increased after SDS treatment for all the conditions tested. However, a decreasing trend in the A_{570}/A_{600} ratio with increasing concentrations of CFCS could suggest significant detachment of biofilm cells. Our results have clearly proved that the Congo red binding assay suggested a significant decrement of EPS after treating with 1% and 2% v/v dosages of *B. casei* CFCS. The Congo red assay indicated a reduction in slime layer formation after *B. casei* CFCS treatment, so it could subsequently enhance the penetrating capacity of a therapeutic drug through a resistance modifying action [68]. Equivalent results were obtained by Parai et al. [69], the amount of Congo red in the supernatant of *P. aeruginosa* was significantly decreased from 100% (in the vehicle control) to 60% and 30% in the IC₅₀ and IC₈₀ reserpine treatment set, respectively.

Rhamnolipids are bio-surfactants produced by *P. aeruginosa* that have a role in swarming motility and biofilm architecture [39]. The present study depicted that 1% and 2% v/v of *B. casei* CFCS significantly decreased rhamnolipid production by 10.52% and 12.48%, respectively (Fig. 3D). The enhanced cell hydrophobicity is associated with a

reduction of the total cellular LPS content. AlTahhan et al. [70], observed a loss of LPS in *P. aeruginosa* strains treated with rhamnolipid at low concentrations resulted in increased cell surface hydrophobicity. Rhamnolipids are reported to stimulate motility, thereby inhibiting attachment by preventing cells from adhering tightly to the substratum. During biofilm development, *rhl* pathway is responsible for the production of amphiphilic biosurfactants known as rhamnolipids are stated to create and maintain fluid channels for water and oxygen flow around the base of the biofilm [71]. Our study clearly showed a significant decrease in rhamnolipid production of *P. aeruginosa* treated with *B. casei* CFCS are in accordance with the previous study of Packiavathy et al. [72], and Abraham et al., [32].

3.6. Pellicle formation assay

Pellicle formation (thin biofilm surrounding cells that assembles at the air-liquid interface of a standing liquid culture) is encoded by the genes *pelA-G* in *P. aeruginosa* [40]. Transcription of the exopolysaccharide-related *pelA-G* gene loci is controlled indirectly by *las* and *rhl* QS systems however this process is as yet largely undefined. *P. aeruginosa* can form robust pellicles when grown in standing culture. In this experiment, the untreated control tubes showed the formation of robust pellicle with greenish pigments, but it visibly decreased after 1% v/v CFCS treatments (Fig. 4A). Recently, Parai et al. [69], reported that reserpine reduced the pellicle formation in standing liquid culture of *P. aeruginosa* PAO1 in LB broth at IC₅₀ and IC₈₀ concentrations, respectively. Our study predicted that *B. casei* CFCS could also restrict pellicle formation at the air-liquid interface and hinder bacterial attachment and growth as a biofilm. The ability of a compound to destroy or inhibit these characteristics could make it a good antibiofilm agent.

3.7. Swarming assay

Bacterial motility is crucial in the effort of the microbes to reach the substratum. Once attached to the surface, the bacteria spread all around via swarming and swimming type of motility, ultimately leading to the

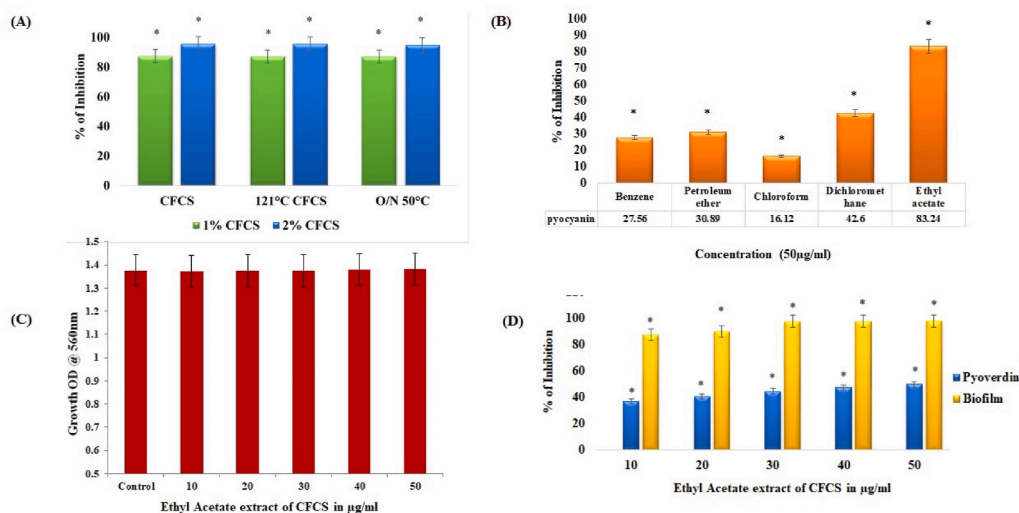


Fig. 5. (A) Analysis of the thermal stability of *B. casei* CFCS through *P. aeruginosa* pyocyanin inhibition assay, (B) Effect of the successive solvent extracts of *B. casei* CFCS on *P. aeruginosa* pyocyanin production, (C) Effect of the *B. casei* CFCS ethyl acetate extract on *P. aeruginosa* growth, and (D) pyoverdinin and biofilm production. Data represent mean values ($n = 3$) $\pm$ standard deviation. * indicates significance at $P < 0.005$.

severity and persistence of biofilm formation over the surface [73]. The present investigation reported that the effect of *B. casei* CFCS visibly inhibited swarming motility of *P. aeruginosa* at a concentration of 1% and 2% v/v (Fig. 4B). The CFCS reduced flagellum driven motility of *P. aeruginosa* in the treated sample compared to the control. Thus, bacterial flagella, fimbriae and pili might play some important roles during biofilm development [74]. According to Singh et al. [62], the swarming motility of PAO1 and PAH are significantly inhibited in the presence of *Exiguobacterium indicum* SJ16 extract isolated from the rhizosphere of *Cyperus laevigatus* in the plate assay. Our study also shows that the CFCS of *B. casei* visibly inhibited swarming motility of *P. aeruginosa* because of the capability to block the initial attachment of *P. aeruginosa* by preventing the bacterial motility. This strategy could potentially open new avenue in the effort to halt bacterial spreading and thus minimize the ability of microbes to form biofilms.

3.8. Thermal stability of *B. casei* CFCS

CFCS of *B. casei* was subjected to heat treatment. The obtained result shows that pyocyanin production was inhibited in *P. aeruginosa* on treatment with *B. casei* CFCS heat inactivated even at 121 °C and overnight heating at 50 °C (Fig. 5A). So, the *B. casei* CFCS was found to be a thermostable molecule, probably an analogue compound and not an AHL-degrading enzyme. A similar result was obtained by Devi et al. [45], that the heat inactivated even at 100 °C CFCS of *Bacillus subtilis* R-18 inhibited prodigiosin production of *S. marcescens*. Hence, in this study the result of heat treatment suggested that the constituents present in *B. casei* CFCS would be an analogue compound. Thus, *B. casei* CFCS was subjected to solvent extraction.

3.9. Solvent extraction

In order to determine the active principle, CFCS was collected from 24-h grown *B. casei* culture and subjected to polarity-based extraction. The organic phases of each extract were concentrated under reduced pressure and screened through pyocyanin assay. In addition, among the various solvents extracts of *B. casei* tested, ethyl acetate extract showed significant inhibition in pyocyanin pigment production in *P. aeruginosa*. Padmavathi et al. [61], also found that the various solvent extracts tested, chloroform extract of *Vibrio alginolyticus* (G16) showed an 80% reduction in the production of prodigiosin in *S. marcescens*. Similar findings of Rashiya et al. [24], found that the acetone extract of marine

sponge *Leucosolinia eleanor* shows higher antimicrobial activity against *E. coli*, *Pseudomonas aeruginosa* and *Candida albicans*. Among the various solvent extracts tested, ethyl acetate (50 µg/ml *B. casei* CFCS) extract showed 83.24% inhibition in the pyocyanin pigment production in *P. aeruginosa* (Fig. 5B) and, hence, this extract alone was chosen for further study.

3.10. Effect of ethyl acetate extract of *B. casei* CFCS on pyoverdinin and biofilm of *P. aeruginosa*

In the present study, ethyl acetate extract of *B. casei* CFCS (10–50 µg/ml) showed potential inhibitory activity against *P. aeruginosa* pyoverdinin and biofilm production without affecting the planktonic growth of *P. aeruginosa* (Fig. 5C and D). In Fig. 5D, the production of pyoverdinin was significantly decreased to the level of 37% and 50% at 10 µg/ml and 50 µg/ml concentrations, respectively. Also, the ethyl acetate extract of *B. casei* CFCS exhibited a significant antibiofilm activity against *P. aeruginosa*, wherein, ethyl acetate extract of *B. casei* CFCS shows 88% and 98% of biofilm inhibition at 10 µg/ml and 50 µg/ml concentrations, respectively (Fig. 5D).

P. aeruginosa exhibits an ample biofilm-forming behaviour, which is one of the most important requisites for its pathogenicity [10]. *P. aeruginosa* has two well-studied QS systems, *las* and *rhl*. The *las* system is comprised of the transcriptional activator *LasR* and the autoinducer synthase enzyme *LasI*, which directs synthesis of the signal molecule *N*-(3-oxododecanoyl) homoserine lactone (3O-C₁₂-HSL). In studies examining QS and *P. aeruginosa* biofilm formation, a PAO1 mutant strain deficient in production of the *las* signal molecule, 3O-C₁₂-HSL, formed very thin biofilms that lacked the three-dimensional architecture observed with the parent. Similarly, the *rhl* system is comprised of the transcriptional activator *RhlR* and the *RhlI* autoinducer synthase that synthesizes *N*-butyryl homoserine lactone (C₄-HSL) [10]. *LasI* and *LasR* QS systems are important to regulate the production of virulence factors and biofilm formation in *P. aeruginosa* [75]. Since, both *lasI* and *rhlI* genes are autoregulated; *lasI* requires 3O-C₁₂-HSL, and *rhlI* depends on the presence of both 3O-C₁₂-HSL and C₄-HSL for maximal expression.

One of the main difficulties in treating biofilm infections arises from a lack of understanding of the phenotypic characteristics of the entire growth cycle of a biofilm-forming bacterial pathogen and also requires knowledge of whether the bacterial population is of one phenotype or of many to treat biofilm infections [76]. Biofilms confer superior survival capability to the bacteria by providing a physical barrier against the

Table 1

List of chemical constituents identified from *B. casei* CFCS ethyl acetate extract through GC-MS analysis.

Peak	Retention time	Area %	Compound Name
10	6.009	19.37	Isovaline, 3-hydroxy-
42	19.942	12.53	Diethyl Phthalate
11	6.167	9.33	Butanoic acid, 2-ethyl-
65	26.09	6.4	Pyrrolo [1,2-a]pyrazine-1,4-dione, hexahydro-3-(2-methylpropyl)-
72	27.179	3.2	Hexadecanoic acid, 14-methyl-, methyl ester
51	23.788	2.8	Tetradecanoic acid, 12-methyl-, methyl ester, (S)-
64	26.008	2.62	Hexadecanoic acid, methyl ester
20	10.94	2.13	Benzeneacetic acid, methyl ester
71	27.044	1.63	Hexadecanoic acid, 15-methyl-, methyl ester
54	24.463	2.94	2-dodecenal
69	26.599	1.62	n-Hexadecanoic acid
62	25.773	1.48	Pyrrolo [1,2-a]pyrazine-1,4-dione, hexahydro-3-(2-methylpropyl)-

entry of antimicrobial agents, thereby developing resistance to antibiotics. Given the widespread occurrence of AHL-mediated cell-to-cell communication systems, it has been revealed that interfering with the QS system of *P. aeruginosa* may pave the way to prevent the development of biofilm and onset of infections. Another important event in the process of biofilm development is its characteristic architecture, which is mainly dependent on EPS production [77]. It was well proven that the production of EPS is also dependent on AHL [78], and therefore, it is envisaged that the inhibition of AHL can possibly prevent the expression of EPS synthesis and lead to the loose architecture of the *P. aeruginosa* biofilm is probably related to the inhibition of the QS system by *B. casei*, as compared with that of the control.

3.11. Characterization of bioactive fraction by GC-MS analysis

GC-MS analysis showed a total of 12 different compounds from *B. casei* CFCS ethyl acetate extract, as listed in Table 1 and chromatogram in Fig. 6 in which Diethyl phthalate (DEP) accounted for major proportions of up to 12.53% with retention time of 19 min. The di-(2-ethylhexyl) phthalate and dibutyl phthalate from *Brevibacterium mcbrellneri* have already reported for broad spectrum antibacterial activity and mosquito larvicidal activity [79]. Similarly, DEP has been

stated from various fungal and algal sources with antimicrobial and antibiofilm properties have previously been reported for its ability to interfere with QS-mediated virulence factors production and biofilm formation in *P. aeruginosa* [80,81].

Moreover, GC-MS analysis of ethyl acetate extract of *B. casei* CFCS accounted for major proportions of DEP. To substantiate the fact, Varsha Gayathri et al. [81], stated that the first report of antibiofilm activity of DEP at retention time 15.563 and area (39.28%) was identified through GC-MS analysis of ethanolic extract of *Chlorella vulgaris*. The extracts were found to be non-toxic at all the concentrations tested and marks as a valuable target for developing biofilm inhibitors against *P. aeruginosa*. In addition, previous literature evidenced the use of glycolipid bio-surfactant isolated from the marine actinobacterium *B. casei* MSA19 was evaluated for antibiofilm potential against pathogenic biofilms [82]. Recently, Alvertoxins with QS inhibitory activities against *C. violaceum* CV026 from the marine-derived fungus *Cladosporium* sp. KFD33 were reported [83]. Hence, it is envisioned that the presence of DEP in the *B. casei* CFCS ethyl acetate extract may probably affect the QS-mediated virulence factors production and biofilm formation in *P. aeruginosa*.

So far, to the best of our knowledge, *B. casei* CFCS ethyl acetate extract has not yet been explored for its anti-QS and antibiofilm activity against *P. aeruginosa* PAO1. Therefore, this is the first of its kind to report the QSI activity of *B. casei* ethyl acetate extract by possibly targeting the QS signaling pathway of *P. aeruginosa*. As ethyl acetate extract of *B. casei* exhibits potent antivirulence activities, it could be used as an alternative drug of choice to prevent hospital-acquired *P. aeruginosa* infections.

4. Conclusion

In conclusion, the present work delineates that *B. casei* CFCS ethyl acetate extract could inhibit the biofilm formation and virulence factors production such as pyocyanin, pyoverdine, rhamnolipid, EPS and motility of PAO1. Hence, this study highlights that the presence of Diethyl phthalate in *B. casei* ethyl acetate extract could be an efficient anti-virulence agent to treat persistent and drug-resistant *P. aeruginosa* infections.

Author contributions

NR, RD, AR and GS designed the study. NR performed the experiments. NR, NP, AA, PP and NS interpreted the data. AR and GS contributed to materials and reagents. NR drafted the manuscript. GS

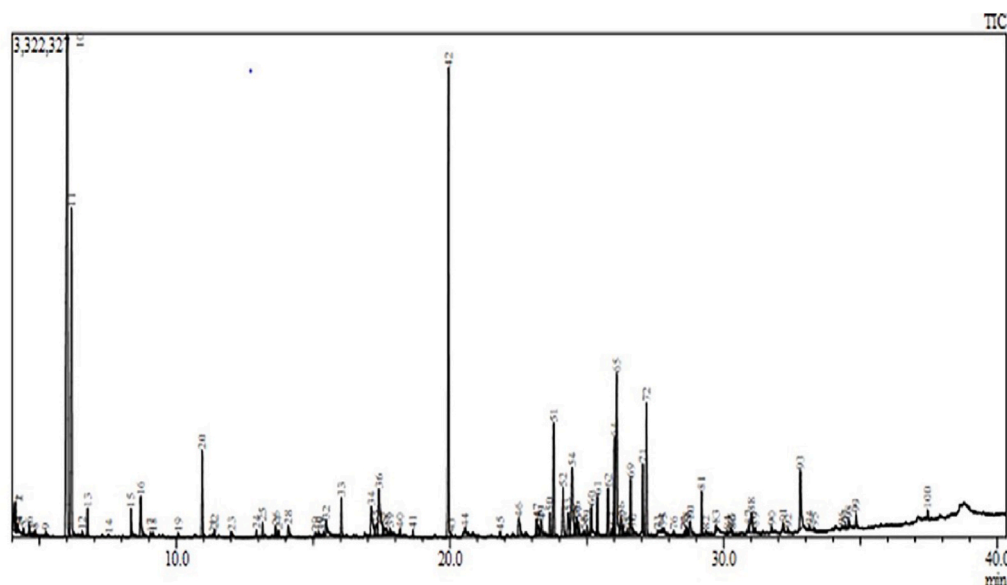


Fig. 6. GC- MS analysis - chromatogram of ethyl acetate extract of *B. casei* CFCS.

and SG corrected the manuscript. All authors approved the final version of the manuscript.

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Declaration of competing interest

The authors declare that there are no conflicts of interest.

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